

Операционные системы

Основы интерфейса взаимодействия пользователя с системой Unix на уровне
командной строки

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Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

1. Цели и задачи работы



Приобретение практических навыков взаимодействия пользователя с системой посредством командной строки.



- 1 Определить имя и путь домашнего каталога
- 2 Изучить команду `ls`.
- 3 Выполнить действия с каталогами.
- 4 Получить дополнительные сведения при помощи справки по командам.
- 5 Изучить команду `history`.

2. Процесс выполнения лабораторной работы



```
irabzalimov@irabzalimov:~$  
irabzalimov@irabzalimov:~$ cd  
irabzalimov@irabzalimov:~$ pwd  
/home/irabzalimov  
irabzalimov@irabzalimov:~$
```

Рисунок 1: Путь к домашнему каталогу



```
irabzalimov@irabzalimov:/tmp$ ls
happd.sock
snap-private-tmp
systemd-private-fbd8f451ac784e2e8221b6fc0a991134-abrt.service-hiEWXn
systemd-private-fbd8f451ac784e2e8221b6fc0a991134-chronyd.service-8IwKi8
systemd-private-fbd8f451ac784e2e8221b6fc0a991134-colord.service-iTkPEv
systemd-private-fbd8f451ac784e2e8221b6fc0a991134-dbus-broker.service-UtFX9A
systemd-private-fbd8f451ac784e2e8221b6fc0a991134-fwupd.service-7VyA6Q
systemd-private-fbd8f451ac784e2e8221b6fc0a991134-irqbalance.service-MCQWV8
systemd-private-fbd8f451ac784e2e8221b6fc0a991134-low-memory-monitor.service-YiVLhZ
systemd-private-fbd8f451ac784e2e8221b6fc0a991134-ModemManager.service-jV4JbB
systemd-private-fbd8f451ac784e2e8221b6fc0a991134-pcsd.service-D9oLNs
systemd-private-fbd8f451ac784e2e8221b6fc0a991134-polkit.service-VffmM0
systemd-private-fbd8f451ac784e2e8221b6fc0a991134-rtkit-daemon.service-9Wl9CV
systemd-private-fbd8f451ac784e2e8221b6fc0a991134-switcheroo-control.service-WXRIJ3
systemd-private-fbd8f451ac784e2e8221b6fc0a991134-systemd-logind.service-a77rht
systemd-private-fbd8f451ac784e2e8221b6fc0a991134-systemd-resolved.service-GyKZZH
systemd-private-fbd8f451ac784e2e8221b6fc0a991134-upower.service-B9Aj5u
vmware-root_839-3979774022
irabzalimov@irabzalimov:/tmp$
```

Рисунок 2: Команда ls



```
irabzalimov@irabzalimov:/tmp$ ls -a
.
..
.font-unix
happd.sock
.ICE-unix
snap-private-tmp
systemd-private-fbd8f451ac784e2e8221b6fc0a991134-abrt.service-hiEWXn
systemd-private-fbd8f451ac784e2e8221b6fc0a991134-chronyd.service-8IwKi8
systemd-private-fbd8f451ac784e2e8221b6fc0a991134-colord.service-iTkPEv
systemd-private-fbd8f451ac784e2e8221b6fc0a991134-dbus-broker.service-UtFX9A
systemd-private-fbd8f451ac784e2e8221b6fc0a991134-fwupd.service-7VyAGQ
systemd-private-fbd8f451ac784e2e8221b6fc0a991134-irqbalance.service-MCQWV8
systemd-private-fbd8f451ac784e2e8221b6fc0a991134-low-memory-monitor.service-YiVLhZ
systemd-private-fbd8f451ac784e2e8221b6fc0a991134-ModemManager.service-jV4JbB
systemd-private-fbd8f451ac784e2e8221b6fc0a991134-pcsd.service-D9oLNs
systemd-private-fbd8f451ac784e2e8221b6fc0a991134-polkit.service-VfFmM0
systemd-private-fbd8f451ac784e2e8221b6fc0a991134-rtkit-daemon.service-9Wl9CV
systemd-private-fbd8f451ac784e2e8221b6fc0a991134-switcheroo-control.service-WXRIJ3
systemd-private-fbd8f451ac784e2e8221b6fc0a991134-systemd-logind.service-a77rht
systemd-private-fbd8f451ac784e2e8221b6fc0a991134-systemd-resolved.service-6yKZZH
systemd-private-fbd8f451ac784e2e8221b6fc0a991134-upower.service-B9Aj5u
vmware-root_839-3979774022
.X0-lock
.X11-unix
.X1-lock
.XIM-unix
irabzalimov@irabzalimov:/tmp$
```

Рисунок 3: Команда ls -a



```
irabzalimov@irabzalimov:/tmp$ ls -l
total 0
srw-rw-rw-. 1 root root 0 Oct  2 18:00 happd.sock
drwx-----, 2 root root 40 Oct  2 18:00 snap-private-tmp
drwx-----, 3 root root 60 Oct  2 18:00 systemd-private-fbd8f451ac784e2e8221b6fc0a991134-abr
td.service-hiEWXn
drwx-----, 3 root root 60 Oct  2 18:00 systemd-private-fbd8f451ac784e2e8221b6fc0a991134-chr
onyd.service-8lwKi8
drwx-----, 3 root root 60 Oct  2 18:01 systemd-private-fbd8f451ac784e2e8221b6fc0a991134-col
ord.service-iTkPEv
drwx-----, 3 root root 60 Oct  2 18:00 systemd-private-fbd8f451ac784e2e8221b6fc0a991134-dbu
s-broker.service-UtFX9A
drwx-----, 3 root root 60 Oct  2 18:01 systemd-private-fbd8f451ac784e2e8221b6fc0a991134-fwu
pd.service-7VyA6Q
drwx-----, 3 root root 60 Oct  2 18:00 systemd-private-fbd8f451ac784e2e8221b6fc0a991134-irq
balance.service-MCQWV8
drwx-----, 3 root root 60 Oct  2 18:00 systemd-private-fbd8f451ac784e2e8221b6fc0a991134-low
-memory-monitor.service-YiVLhZ
drwx-----, 3 root root 60 Oct  2 18:00 systemd-private-fbd8f451ac784e2e8221b6fc0a991134-Mod
emManager.service-jV4JbB
drwx-----, 3 root root 60 Oct  2 18:00 systemd-private-fbd8f451ac784e2e8221b6fc0a991134-pcs
cd.service-D9oLNs
drwx-----, 3 root root 60 Oct  2 18:00 systemd-private-fbd8f451ac784e2e8221b6fc0a991134-pol
kit.service-VfFmM0
drwx-----, 3 root root 60 Oct  2 18:00 systemd-private-fbd8f451ac784e2e8221b6fc0a991134-rtk
it-daemon.service-9Wl9CV
drwx-----, 3 root root 60 Oct  2 18:00 systemd-private-fbd8f451ac784e2e8221b6fc0a991134-swi
tcheroo-control.service-WXRIJ3
drwx-----, 3 root root 60 Oct  2 18:00 systemd-private-fbd8f451ac784e2e8221b6fc0a991134-sys
```

Рисунок 4: Команда ls -l



```
irabzalimov@irabzalimov:/tmp$ ls -f
..
systemd-private-fbd8f451ac784e2e8221b6fc0a991134-fwupd.service-7VyA6Q
.X1-lock
.X0-lock
systemd-private-fbd8f451ac784e2e8221b6fc0a991134-pcsd.service-D9oLNs
systemd-private-fbd8f451ac784e2e8221b6fc0a991134-colord.service-iTkPEv
happd.sock
systemd-private-fbd8f451ac784e2e8221b6fc0a991134-ModemManager.service-jV4JbB
vmware-root_839-3979774022
systemd-private-fbd8f451ac784e2e8221b6fc0a991134-systemd-logind.service-a77rht
systemd-private-fbd8f451ac784e2e8221b6fc0a991134-upower.service-B9Aj5u
systemd-private-fbd8f451ac784e2e8221b6fc0a991134-switcheroo-control.service-WXRIJ3
systemd-private-fbd8f451ac784e2e8221b6fc0a991134-rtkit-daemon.service-9WL9CV
systemd-private-fbd8f451ac784e2e8221b6fc0a991134-polkit.service-VfFmM0
systemd-private-fbd8f451ac784e2e8221b6fc0a991134-low-memory-monitor.service-YiVLhZ
systemd-private-fbd8f451ac784e2e8221b6fc0a991134-chronyd.service-8IwKi8
systemd-private-fbd8f451ac784e2e8221b6fc0a991134-irqbalance.service-MCQWV8
systemd-private-fbd8f451ac784e2e8221b6fc0a991134-abrt.service-hiEWXn
systemd-private-fbd8f451ac784e2e8221b6fc0a991134-dbus-broker.service-UtFX9A
systemd-private-fbd8f451ac784e2e8221b6fc0a991134-systemd-resolved.service-GyKZZH
.font-unix
.XIM-unix
.ICE-unix
.X11-unix
snap-private-tmp
irabzalimov@irabzalimov:/tmp$
```

Рисунок 5: Команда ls -f



```
irabzalimov@irabzalimov:~/tmp$  
irabzalimov@irabzalimov:~/tmp$ cd /var/spool/  
irabzalimov@irabzalimov:/var/spool$ ls -l  
total 0  
drwxr-x--x. 1 root abrt 24680 Aug  3 03:00 abrt  
drwx-----. 1 abrt abrt    0 Aug  3 03:00 abrt-upload  
drwxr-xr-x. 1 root root   66 Oct 23  2025 anacron  
drwx-----. 1 root root   18 Oct 23  2025 at  
drwx-----. 1 root root    0 Aug  6  2025 cron  
drwx--x---. 1 root lp     6 May 26 03:00 cups  
drwxr-xr-x. 1 root root    0 Jul 30  2025 lpd  
drwxrwxr-x. 1 root mail  152 Oct  2 16:06 mail  
drwxr-xr-x. 1 root root    0 Jan 27  2026 plymouth  
irabzalimov@irabzalimov:/var/spool$
```

Рисунок 6: Каталог /var/spool



```
irabzalimov@irabzalimov:/var/spool$ cd
irabzalimov@irabzalimov:~$ ls
Desktop  Downloads  Music      Public      Videos
Documents git-extended Pictures  Templates  work
irabzalimov@irabzalimov:~$ ls -al
total 28
drwx-----. 1 irabzalimov irabzalimov 370 Oct 2 17:06 .
drwxr-xr-x. 1 root root 152 Oct 2 16:06 ..
-rw-----. 1 irabzalimov irabzalimov 4639 Oct 2 17:14 .bash_history
-rw-r--r--. 1 irabzalimov irabzalimov 18 Jul 23 2025 .bash_logout
-rw-r--r--. 1 irabzalimov irabzalimov 144 Jul 23 2025 .bash_profile
-rw-r--r--. 1 irabzalimov irabzalimov 681 Oct 2 16:48 .bashrc
drwx-----. 1 irabzalimov irabzalimov 498 Oct 2 18:02 .cache
drwx-----. 1 irabzalimov irabzalimov 360 Oct 2 16:53 .config
drwxr-xr-x. 1 irabzalimov irabzalimov 0 Oct 2 16:09 Desktop
drwxr-xr-x. 1 irabzalimov irabzalimov 0 Oct 2 16:09 Documents
drwxr-xr-x. 1 irabzalimov irabzalimov 0 Oct 2 16:09 Downloads
-rw-r--r--. 1 irabzalimov irabzalimov 334 Aug 28 03:00 .emacs
-rw-r--r--. 1 irabzalimov irabzalimov 236 Oct 2 17:06 .gitconfig
drwxr-xr-x. 1 irabzalimov irabzalimov 74 Oct 2 16:59 git-extended
drwx-----. 1 irabzalimov irabzalimov 136 Oct 2 16:23 .gnupg
drwx-----. 1 irabzalimov irabzalimov 20 Oct 2 16:09 .local
drwxr-xr-x. 1 irabzalimov irabzalimov 34 Oct 23 2025 .mozilla
drwxr-xr-x. 1 irabzalimov irabzalimov 0 Oct 2 16:09 Music
drwxr-xr-x. 1 irabzalimov irabzalimov 0 Oct 2 16:09 Pictures
drwxr-xr-x. 1 irabzalimov irabzalimov 0 Oct 2 16:09 Public
drwx-----. 1 irabzalimov irabzalimov 132 Oct 2 16:27 .ssh
drwxr-xr-x. 1 irabzalimov irabzalimov 0 Oct 2 16:09 Templates
drwxr-xr-x. 1 irabzalimov irabzalimov 18 Oct 2 16:31 .texlive2025
drwxr-xr-x. 1 irabzalimov irabzalimov 0 Oct 2 16:09 Videos
drwxr-xr-x. 1 irabzalimov irabzalimov 10 Oct 2 16:26 work
irabzalimov@irabzalimov:~$
```

Рисунок 7: Файлы в домашнем каталоге



```
irabzalimov@irabzalimov:~$ mkdir newdir
irabzalimov@irabzalimov:~$ mkdir newdir/morefun
irabzalimov@irabzalimov:~$ mkdir letters memos misk
irabzalimov@irabzalimov:~$ ls
Desktop  Downloads  letters  misk  newdir  Public  Videos
Documents  git-extended  memos  Music  Pictures  Templates  work
irabzalimov@irabzalimov:~$ rm letters/ memos/ misk/
rm: cannot remove 'letters/': Is a directory
rm: cannot remove 'memos/': Is a directory
rm: cannot remove 'misk/': Is a directory
irabzalimov@irabzalimov:~$ rm -r letters/ memos/ misk/
irabzalimov@irabzalimov:~$ rm -r newdir/
irabzalimov@irabzalimov:~$ ls
Desktop  Documents  Downloads  git-extended  Music  Pictures  Public  Templates  Videos  work
irabzalimov@irabzalimov:~$
```

Рисунок 8: Действия с каталогами



```
./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/report/_extensions/yamadharma/minted-quarto':
_extension.yml  minted-quarto.lua

./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/report/image':
solvay.jpg

./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/report/_resources':
csl  tex

./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/report/_resources/csl':
gost-r-7-0-5-2008-numeric.csl

./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/report/_resources/tex':
preamble.tex

./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/scripts':
image-report mpv-shot
irabzalimov@irabzalimov:~$
irabzalimov@irabzalimov:~$ ls -t
git-extended work Desktop Documents Downloads Music Pictures Public Templates Videos
irabzalimov@irabzalimov:~$
```

Рисунок 9: Команда ls -R и ls -t



```
irabzalimov@irabzalimov:~$ help cd
cd: cd [-L|[-P [-e]] [-@]] [dir]
    Change the shell working directory.

Change the current directory to DIR.  The default DIR is the value of the
HOME shell variable.  If DIR is "-", it is converted to $OLDPWD.

The variable CDPATH defines the search path for the directory containing
DIR.  Alternative directory names in CDPATH are separated by a colon (:).
A null directory name is the same as the current directory.  If DIR begins
with a slash (/), then CDPATH is not used.

If the directory is not found, and the shell option `cdable_vars' is set,
the word is assumed to be a variable name.  If that variable has a value,
its value is used for DIR.

Options:
  -L      force symbolic links to be followed: resolve symbolic
          links in DIR after processing instances of `..'
  -P      use the physical directory structure without following
          symbolic links: resolve symbolic links in DIR before
          processing instances of `..'
  -e      if the -P option is supplied, and the current working
          directory cannot be determined successfully, exit with
          a non-zero status
  -@      on systems that support it, present a file with extended
          attributes as a directory containing the file attributes

The default is to follow symbolic links, as if `-L' were specified.
`..' is processed by removing the immediately previous pathname component
back to a slash or the beginning of DIR.
```

Рисунок 10: Справка по команде cd



```
PWD(1)                                User Commands                                PWD(1)

NAME
    pwd - print name of current/working directory

SYNOPSIS
    pwd [OPTION]...

DESCRIPTION
    Print the full filename of the current working directory.

    -L, --logical
        use PWD from environment, even if it contains symlinks

    -P, --physical
        resolve all symlinks

    --help display this help and exit

    --version
        output version information and exit

    If no option is specified, -P is assumed.

    Your shell may have its own version of pwd, which usually supersedes the version described
    here. Please refer to your shell's documentation for details about the options it sup-
    ports.

AUTHOR
    Written by Jim Meyering.

REPORTING BUGS
    Manual page pwd(1) line 1 (press h for help or q to quit)
```

Рисунок 11: Справка по команде pwd



```
MKDIR(1)                                User Commands                                MKDIR(1)

NAME
    mkdir - make directories

SYNOPSIS
    mkdir [OPTION]... DIRECTORY...

DESCRIPTION
    Create the DIRECTORY(ies), if they do not already exist.

    Mandatory arguments to long options are mandatory for short options too.

    -m, --mode=MODE
        set file mode (as in chmod), not a=rwx - umask

    -p, --parents
        no error if existing, make parent directories as needed, with their file modes un-
        affected by any -m option

    -v, --verbose
        print a message for each created directory

    -Z
        set SELinux security context of each created directory to the default type

    --context[=CTX]
        like -Z, or if CTX is specified then set the SELinux or SMACK security context to
        CTX

    --help display this help and exit

    --version

Manual page mkdir(1) line 1 (press h for help or q to quit)
```

Рисунок 12: Справка по команде mkdir



```
RMDIR(1)                                User Commands                                RMDIR(1)

NAME
    rmdir - remove empty directories

SYNOPSIS
    rmdir [OPTION]... DIRECTORY...

DESCRIPTION
    Remove the DIRECTORY(ies), if they are empty.

    --ignore-fail-on-non-empty
        ignore each failure to remove a non-empty directory

    -p, --parents
        remove DIRECTORY and its ancestors; e.g., 'rmdir -p a/b' is similar to 'rmdir a/b
        a'

    -v, --verbose
        output a diagnostic for every directory processed

    --help display this help and exit

    --version
        output version information and exit

AUTHOR
    Written by David MacKenzie.

REPORTING BUGS
    GNU coreutils online help: <https://www.gnu.org/software/coreutils/>
    Report any translation bugs to <https://translationproject.org/team/>
    Manual page rmdir(1) line 1 (press h for help or q to quit)
```

Рисунок 13: Справка по команде rmdir



```
RM(1)                                User Commands                                RM(1)

NAME
  rm - remove files or directories

SYNOPSIS
  rm [OPTION]... [FILE]...

DESCRIPTION
  This manual page documents the GNU version of rm.  rm removes each specified file.  By default, it does not remove directories.

  If the -I or --interactive=once option is given, and there are more than three files or the -r, -R, or --recursive are given, then rm prompts the user for whether to proceed with the entire operation.  If the response is not affirmative, the entire command is aborted.

  Otherwise, if a file is unwritable, standard input is a terminal, and the -f or --force option is not given, or the -i or --interactive=always option is given, rm prompts the user for whether to remove the file.  If the response is not affirmative, the file is skipped.

OPTIONS
  Remove (unlink) the FILE(s).

  -f, --force
    ignore nonexistent files and arguments, never prompt

  -i
    prompt before every removal

  -I
    prompt once before removing more than three files, or when removing recursively; less intrusive than -i, while still giving protection against most mistakes

Manual page rm(1) line 1 (press h for help or q to quit)
```

Рисунок 14: Справка по команде rm



```
198 git add .
199 git commit -am 'upload lab'
200 git push
201 git pull
202 git add .
203 git commit -am 'upload lab'
204 git push
205 cd
206 pwd
207 cd /tmp
208 ls
209 ls -a
210 ls -l
211 ls -f
212 cd /var/spool/
213 ls -l
214 cd
215 ls
216 ls -al
217 mkdir newdir
218 mkdir newdir/morefun
219 mkdir letters memos misk
220 ls
221 rm letters/ memos/ misk/
222 rm -r letters/ memos/ misk/
223 rm -r newdir/
224 ls
225 ls -R
```

Рисунок 15: Команда history

3. Выводы по проделанной работе



Мы приобрели практические навыки взаимодействия пользователя с системой посредством командной строки.